



FINAL PROJECT DOCUMENTATION

Academic Year: 2025-26

Cohort: Jensen Huang

Subject Name : Entrepreneurship

Project Title: WeSafe

Tagline: Your Safety Guardian

Project Type: AI Based Women Safety System

Group Details

Sr No	Student Name	Roll No	Contribution
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1. Problem Statement

Personal safety has become a major concern in today's fast-moving world, especially for women, students, working professionals, and senior citizens. Many existing safety applications require users to manually press an SOS button or unlock their phones during emergencies, which may not be possible in panic situations.

Another challenge is that some safety services stop working when subscriptions expire, creating a risk when protection is needed the most. In addition, hardware-based trackers often face issues such as battery drain, device damage, or poor network connectivity.

These limitations highlight the need for a smarter and more reliable safety solution. WeSafe aims to address this problem by providing quick, continuous, and easy-to-use protection through intelligent safety features and emergency support.

2. OBJECTIVES OF WESAFE

- The primary objective of WeSafe is to provide a reliable and intelligent safety solution that can assist users during emergency situations.

The objectives of the project are:

- To improve personal safety through technology.
- To provide quick emergency assistance through SOS alerts.
- To detect panic situations using voice recognition technology.
- To enable automatic location sharing with trusted contacts.
- To build a community-driven safety network.
- To reduce emergency response time.
- To provide a secure and user-friendly platform.
- To promote safety awareness among users.
- To support real-time communication during emergencies.
- To contribute towards building safer communities.

3. TECHNOLOGY STACK

WeSafe has been developed using modern technologies that support real-time communication, location tracking, cloud storage, and emergency response systems. The selected technology stack ensures reliability, scalability, and cross-platform compatibility.

Frontend Development

- **Flutter**
- **Dart**

Flutter was used to develop a responsive and visually appealing user interface that works across multiple platforms from a single codebase.

Backend Services

- Firebase Firestore
- Firebase Authentication
- Firebase Cloud Functions

Firebase provides secure user authentication, cloud database management, and real-time synchronization of data across users.

Artificial Intelligence & Safety Detection

- Speech Recognition APIs
- speech_to_text Package
- Panic Voice Detection Engine
- Keyword Recognition System

These technologies help detect emergency keywords, panic phrases, and voice-based SOS triggers.

Location & Tracking Services

- GPS Location Tracking
- location Package
- Google Maps Integration

These services provide real-time location monitoring and help emergency contacts track the user's position during emergencies.

Communication & Alert System

- SMS Alert System
- Push Notifications
- Emergency Contact Broadcasting

These technologies ensure that emergency alerts are delivered instantly to trusted contacts.

Local Storage & Data Management

- Shared Preferences

Used for storing user preferences, emergency contacts, application settings, and safety configurations locally on the device.

Hardware Integration

- Microphone Access
- Vibration Services
- SMS Services

These features support panic detection, emergency alerts, and user notifications.

3. SYSTEM REQUIREMENTS

Hardware Requirements

- Android Smartphone
- GPS Support
- Internet Connectivity
- Microphone Access

Software Requirements

- Flutter SDK (3.0+)
- Android Studio
- Firebase Project Setup
- Dart SDK

Permissions Required

The application requires the following permissions for proper functionality:

- Microphone Permission
- Location Permission
- SMS Permission
- Phone Permission
- Contacts Permission

These permissions enable voice detection, emergency alerts, live location sharing, and communication with emergency contacts.

4. KEY FEATURES OF WESAFE

WeSafe provides multiple safety features that work together to offer quick assistance during emergency situations.

1. Voice Trigger SOS

The application continuously monitors emergency keywords and panic phrases. When a trigger word is detected, the system can activate emergency assistance without requiring manual interaction.

2. Live Location Sharing

Users can share their real-time location with trusted emergency contacts, allowing faster response during emergencies.

3. Emergency Contact Alerts

SOS alerts can be instantly sent to pre-configured emergency contacts through SMS and notifications.

4. Community Safety Network

Users can join local safety groups and receive support from nearby community members during emergencies.

5. Smart Route Monitoring

The system monitors travel routes and helps identify unusual movement patterns that may indicate a risky situation.

6. Stealth Protection Mode

The application can operate discreetly through a hidden interface, allowing users to seek help without attracting attention.

7. Silent Evidence Shield

Important information such as location details and emergency records can be stored securely to support future reporting if required.

8. Nearby Safe Spots Directory

Users can quickly locate nearby police stations, hospitals, pharmacies, and transportation facilities.

9. Multi-Language Support

The application supports emergency trigger words in multiple languages to improve accessibility for different users.

5. APPLICATION WORKFLOW

The workflow of WeSafe is designed to provide quick and reliable assistance during emergencies.

Step 1: User Registration

The user creates an account and adds trusted emergency contacts.

Step 2: Background Monitoring

The application runs in the background and continuously monitors emergency keywords and safety indicators.

Step 3: Emergency Detection

If a panic phrase, scream, or emergency trigger is detected, the system evaluates the situation.

Step 4: SOS Activation

The application automatically activates the SOS process and prepares emergency alerts.

Step 5: Location Tracking

The user's live GPS location is captured and shared with trusted contacts.

Step 6: Alert Broadcasting

Emergency messages and notifications are sent to emergency contacts and community safety groups.

Step 7: Assistance and Support

Contacts, nearby community members, or emergency services can respond and provide assistance.

Step 8: Safety Confirmation

The user can confirm their safety status after the situation is resolved.

6. Business Model Link

https://docs.google.com/spreadsheets/d/1_7Oqaeg4St6jeaC_lOVf65emrQRXZtL7xg2-RU9xOTM/edit?usp=sharing

7. Github Repo Link

<https://github.com/riddhi-z1465/WeSafe>

8.App Deployment Link

<https://wesafe-5676c.web.app/>

9.Canvas Model Link :

<https://gamma.app/docs/WeSafe-Indias-most-reliable-personal-safety-platform-8smunllv1q2sv4c>

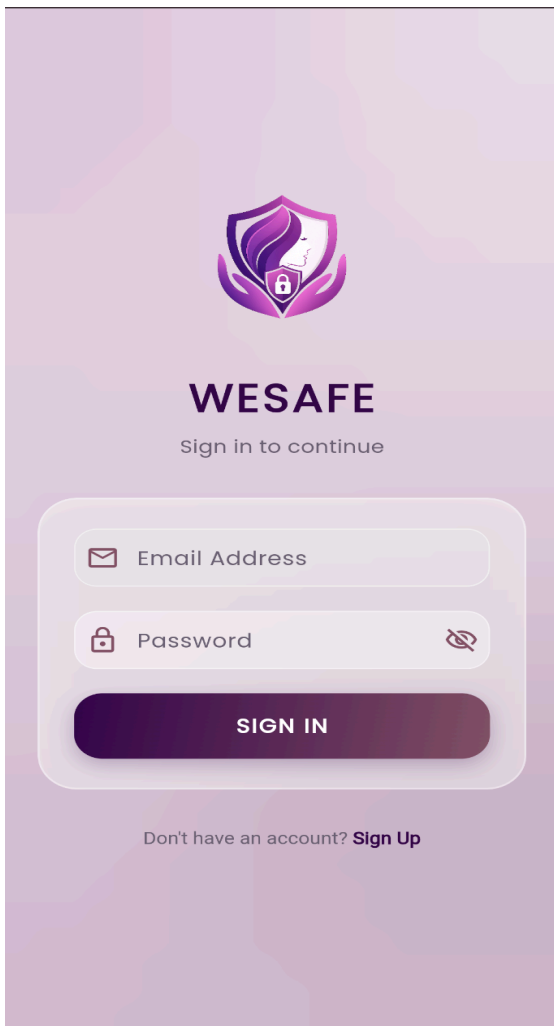
10. FINANCIAL PROJECTION (3 YEARS)

The financial projections are based on expected user growth, premium subscriptions, community adoption, and future institutional partnerships.

https://docs.google.com/spreadsheets/d/1_7Oqaeg4St6jeaC_lOVf65emrQRXZtL7xg2-RU9xOTM/edit?gid=2100846240#gid=2100846240

11. UI / SYSTEM SCREENS

1. Sign in page



The image shows a mobile app sign-in screen for 'WESAFE'. The background is a light purple with a subtle geometric pattern. At the top center is a logo consisting of a shield with a flame-like shape on the left and a profile of a person's head on the right, with a padlock in the center. Below the logo, the word 'WESAFE' is written in a bold, dark purple font. Underneath that, the text 'Sign in to continue' is displayed in a smaller, lighter font. The sign-in form is a rounded rectangle with a light purple background and a subtle drop shadow. It contains two input fields: the first is labeled 'Email Address' with an envelope icon on the left; the second is labeled 'Password' with a padlock icon on the left and an eye icon on the right to toggle visibility. Below these fields is a dark purple button with the text 'SIGN IN' in white, uppercase letters. At the bottom of the form area, the text 'Don't have an account? [Sign Up](#)' is displayed, with 'Sign Up' being a link.



WESAFE

Sign in to continue

 Email Address

 Password 

SIGN IN

Don't have an account? [Sign Up](#)

2. Create account



WESAFE

Create a new account

 Username

 Phone Number

 Email Address

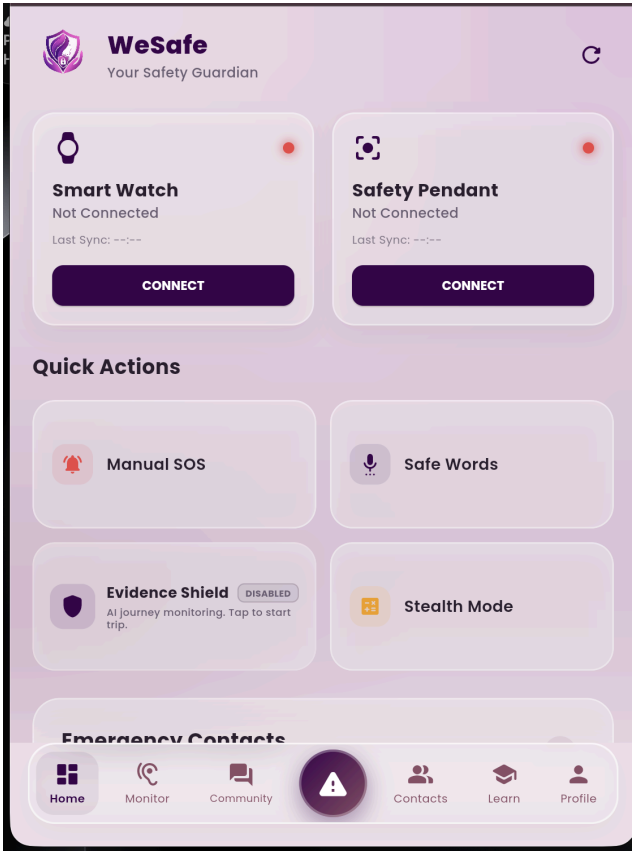
 Password 

 Confirm Password 

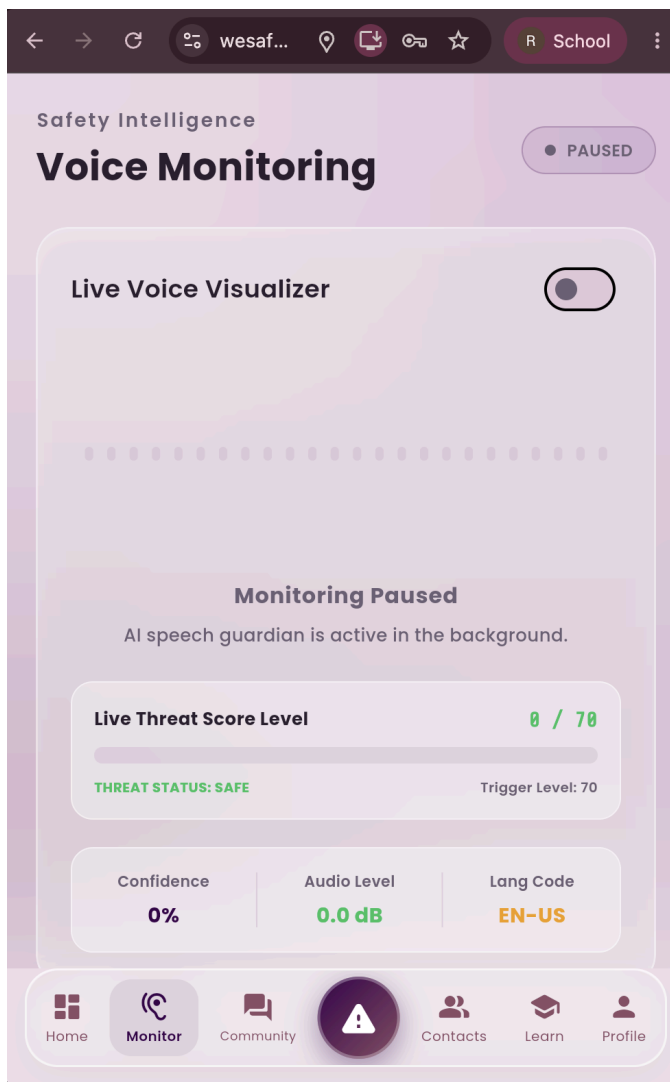
SIGN UP

Already have an account? [Sign In](#)

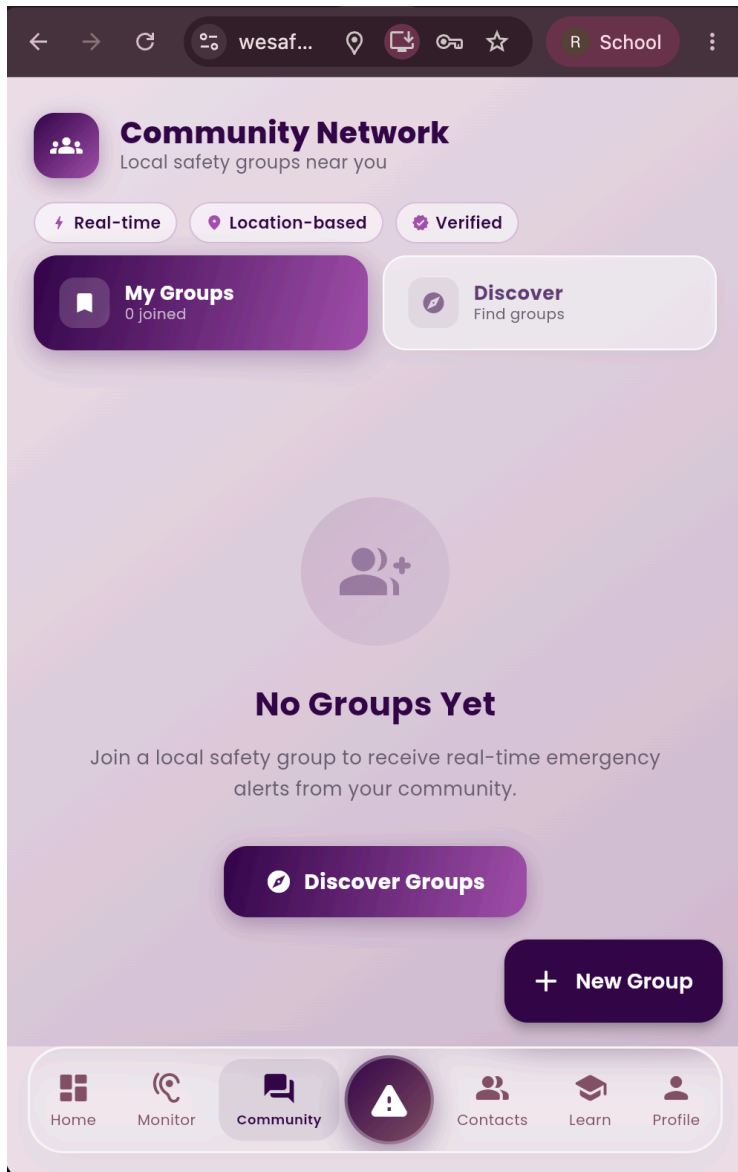
3. Home page



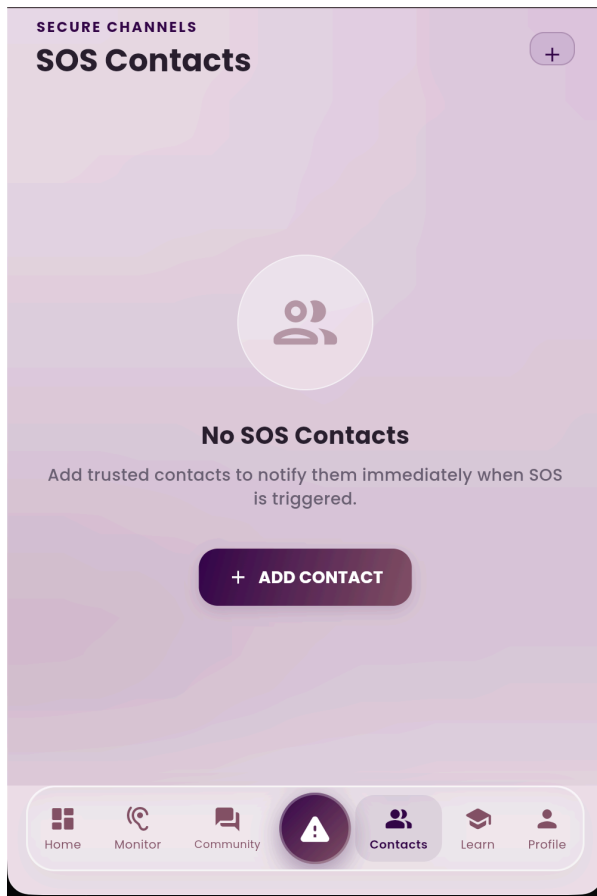
4. Monitor



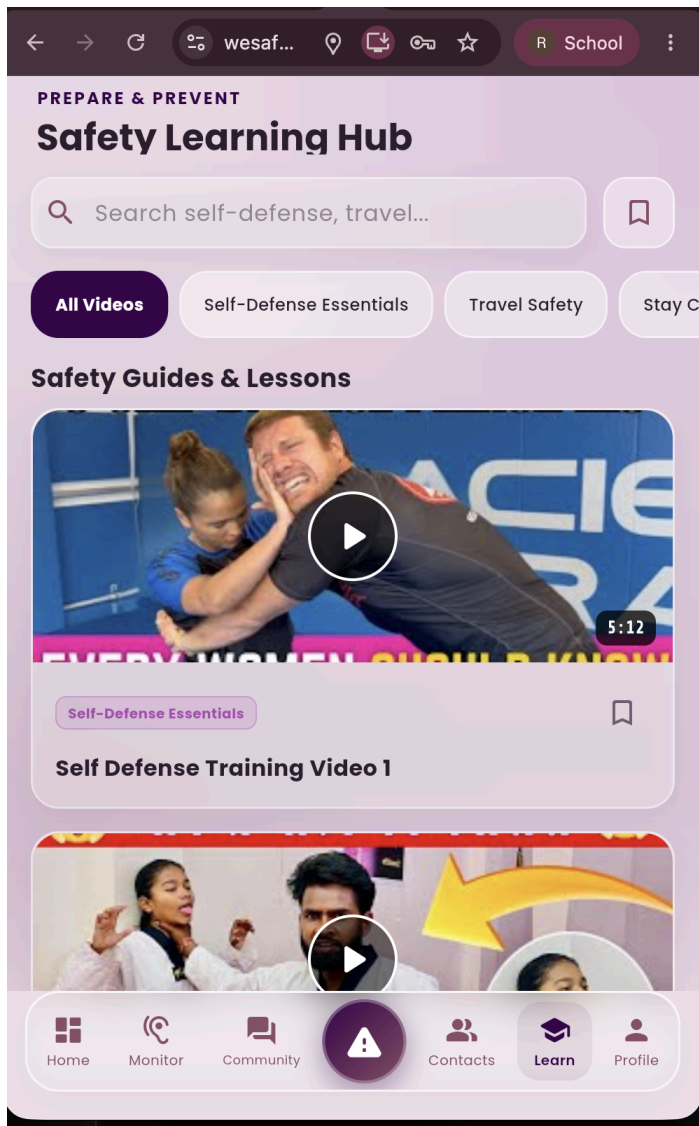
5. Community page



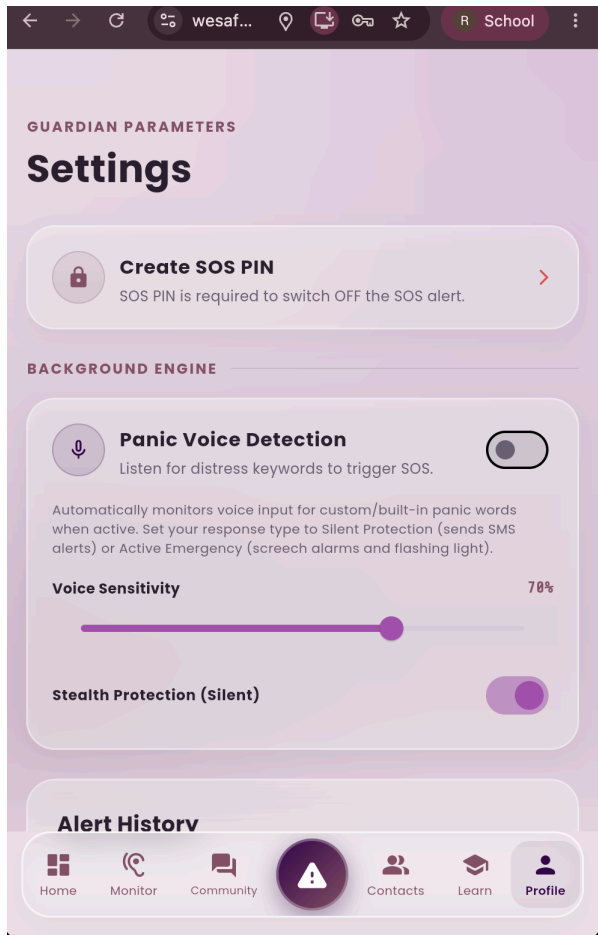
6. Contact page



8. Learn page



8. Profile page



12. FUTURE IMPLEMENTATION AND SCOPE

WeSafe has significant opportunities for future growth and enhancement.

Future Improvements

- Smart wearable device integration.
- AI-based danger prediction and risk analysis.
- Direct police and emergency service integration.
- Advanced community safety features.
- International expansion to support users globally.
- Smart city safety collaborations.
- Enhanced multilingual support.
- Improved AI-powered emergency detection models.

These future developments can further strengthen WeSafe's ability to provide reliable personal safety solutions.

13. CONCLUSION

WeSafe is an AI-based personal safety platform developed to provide quick, reliable, and accessible assistance during emergency situations. By combining voice-triggered SOS activation, live location sharing, community support, and smart monitoring features, the platform aims to improve personal safety and reduce response time during critical situations.

The project demonstrates how modern technologies such as artificial intelligence, cloud services, GPS tracking, and real-time communication can be used to address real-world safety challenges. Beyond individual protection, WeSafe also promotes community participation and safety awareness.

With future enhancements such as wearable integration, advanced AI capabilities, and emergency service connectivity, WeSafe has the potential to become a comprehensive safety ecosystem that contributes to building safer communities and improving public security.

THANKYOU!!!!